

Product Batch - Task Submission - 26.08.2026

ROLL NO : 2303A51027

NAME : Aravind Reddy

1. Salary Band Filtering and Ranking

```
SELECT emp_id, emp_name, salary
FROM employees
WHERE salary BETWEEN 50000 AND 80000
ORDER BY salary DESC, emp_name ASC;
```

2. Multi-Condition Employee Search

```
SELECT emp_id, emp_name, dept_id, salary, manager_id
FROM employees
WHERE (
    (dept_id = 10 AND salary > 55000)
    OR
    (dept_id = 20 AND salary >= 70000)
)
AND manager_id IS NOT NULL;
```

3. Pattern Matching Challenge

```
SELECT emp_id, emp_name
FROM employees
WHERE emp_name LIKE '_%a%'
    AND emp_name NOT LIKE '%n'
ORDER BY emp_name ASC;
```

4. Conditional Salary Classification

```
SELECT

    emp_name,

    salary,

    CASE

        WHEN salary >= 75000 THEN 'High'

        WHEN salary >= 55000 THEN 'Medium'

        ELSE 'Standard'

    END AS salary_band

FROM employees

ORDER BY

    CASE

        WHEN salary >= 75000 THEN 1

        WHEN salary >= 55000 THEN 2

        ELSE 3

    END,

    salary DESC;
```

5. Department Salary Statistics

```
SELECT

    dept_id,

    COUNT(*) AS employee_count,

    MIN(salary) AS min_salary,

    MAX(salary) AS max_salary,

    AVG(salary) AS avg_salary,

    SUM(salary) AS total_salary

FROM employees

GROUP BY dept_id
```

ORDER BY avg_salary DESC;

6. Strong Departments Only

```
SELECT

    dept_id,

    COUNT(*) AS employee_count,

    AVG(salary) AS avg_salary

FROM employees

GROUP BY dept_id

HAVING COUNT(*) >= 2

    AND AVG(salary) > 55000;
```

7. Project Budget Analysis

```
SELECT

    dept_id,

    COUNT(*) AS project_count,

    SUM(budget) AS total_budget

FROM projects

GROUP BY dept_id

HAVING SUM(budget) >= 300000;
```

8. Employee Department Report

```
SELECT

    e.emp_id,

    e.emp_name,

    d.dept_name,

    d.location,

    e.salary

FROM employees e
```

```
INNER JOIN departments d
    ON e.dept_id = d.dept_id
WHERE e.salary >= 60000
ORDER BY d.dept_name ASC, e.salary DESC;
```

9. All Departments With Employee Counts

```
SELECT
    d.dept_id,
    d.dept_name,
    COUNT(e.emp_id) AS employee_count
FROM departments d
LEFT JOIN employees e
    ON d.dept_id = e.dept_id
GROUP BY d.dept_id, d.dept_name;
```

10. Employee and Manager Names

```
SELECT
    e.emp_name AS employee_name,
    e.salary AS employee_salary,
    m.emp_name AS manager_name,
    m.salary AS manager_salary
FROM employees e
INNER JOIN employees m
    ON e.manager_id = m.emp_id
ORDER BY m.emp_name ASC, e.salary DESC;
```